

### UMA021: Numerical Linear Algebra

| L | T | P | Cr  |
|---|---|---|-----|
| 3 | 0 | 2 | 4.0 |

**Course Objectives:** The goal of this course is to give students an introduction to numeric and algorithmic techniques used for the solution of a broad range of mathematical problems, with an emphasis on computational issues and linear algebra. In addition, students will become familiar with numeric programming environments Matlab.

#### Contents:

**Roots of Non-Linear Equations:** Mathematical preliminaries, bisection, fixed-point, Newton's method and its extension to system of equations.

**Interpolation and Integration:** Lagrange and Newton basis of polynomials and interpolation problems, divided difference interpolation, forward and backward differences, trapezoidal and Simpson's rules, method of undetermined coefficients.

**Matrix Algebra:** Gauss elimination method, pivoting strategies, matrix factorization, Jacobi and GaussSeidel methods, matrix norm and conditioning, linear least square problems.

**Matrix Computations:** Orthogonal and orthonormal basis, Gram-Schmidt process, orthogonal matrices and similarity transformations, power method for eigen-value and eigen-vector, QR algorithm, singular value decomposition.

#### Laboratory Work:

Lab experiments will be set in consonance with materials covered in the theory and the implementation of numerical methods will be done using MATLAB

#### Course Learning Outcomes (CLOs) /Course Objectives (COs):

On completion of this course, the students will be able to:

- . Make use of iterative methods to solve nonlinear equations.
- . Approximate the functions using interpolating polynomials and apply to definite integrals.
- . Evaluate solution of system of linear equations and least square problems.
- . Perform matrix computations and evaluate eigen-values and eigen-vectors.

#### Text Books

- . Richard L. Burden, J. Douglas Faires, and Annette Burden, Numerical Analysis, Cengage Learning, 10th edition, 2015.
- . Gilbert Strang, Linear Algebra and its Applications, Cengage Learning, 4th edition, 2005.

- . J. Desmond Higham and Nicholas J. Higham, MATLAB Guide, Third Edition, Society for Industrial and Applied Mathematics, 2016.

### **Reference Books**

- . Steven C. Chapra and Raymond P. Canale, Numerical Methods for Engineers, McGraw-Hill Higher Education, 6th edition, 2010.
- . E. Ward Cheney and David R. Kincaid, Numerical Mathematics and Computing, Cengage Learning, 7th edition, 2012.
- . Endre Suli and David F. Mayers, An Introduction to Numerical Analysis, Cambridge University Press, 2003

### **Evaluation Scheme:**

| <b>Sr. No.</b> | <b>Evaluation elements</b>                      | <b>Weightage (%)</b> |
|----------------|-------------------------------------------------|----------------------|
| 1              | MST                                             | 25                   |
| 2              | EST                                             | 45                   |
| 3              | Sessionals (Assignments/Quizzes/Lab Evaluation) | 30                   |